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The 3-class groups of $\mathbb{Q}(\sqrt[3]{p})$ and its normal closure

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Abstract
We determine the 3-class groups of $\mathbb{Q}(\sqrt[3]{p})$ and $K = \mathbb{Q}(\sqrt[3]{p}, \sqrt{-3})$ when $p \equiv 4, 7 \bmod 9$ is a prime and 3 is a cube modulo p. This confirms a conjecture made by Barrucand-Cohn, and proves the last remaining case of a conjecture of Lemmermeyer on the 3-class group of K.

Keywords Class groups · Pure cubic fields · Ambiguous class number formulas

Mathematics Subject Classification 11R29 · 11R16

1 Introduction

Let p be a prime. Let $F = \mathbb{Q}(\sqrt[3]{p})$ and $K = \mathbb{Q}(\sqrt[3]{p}, \mu_3)$ the normal closure of F. Let A_F (resp. A_K) be the 3-class group (i.e., 3-Sylow subgroup of the class group) of F (resp. K). The paper aims to prove the following result.

Theorem 1.1 *Assume that $p \equiv 4, 7 \mod 9$ is a prime such that the cubic residue symbol $\left(\frac{3}{p}\right)_3 = 1$. Then $A_F \cong \mathbb{Z}/3\mathbb{Z}$ and $A_K \cong (\mathbb{Z}/3\mathbb{Z})^2$.*

This result confirms a conjecture made by Barrucand–Cohn in [2, §8], and later mentioned by Barrucand–Williams–Baniuk, Williams and Gerth in [4, §8, Conjecture 1], [12, p. 273] and [6, p. 474]. Theorem 1.1 also completes a proof of a Lemmermeyer's conjecture on A_K in [8, Conjecture 5, §1.10] when combining with the following known results:

1. If $p \equiv 2 \bmod 3$, then the groups A_F and A_K are both trivial; see [7].
2. If $p \equiv 1 \bmod 3$, then A_F is cyclic non-trivial and $\mathrm{rk}(A_K) = 1$ or 2 where $\mathrm{rk}(A_K)$ is the 3-rank of A_K; see [6].

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3. If $p \equiv 1 \bmod 9$, then $\mathrm{rk}(A_K) = 2$ if and only if 9 divides $|A_F|$; see [5, Lemma 5.11] and [6].
4. If $p \equiv 4, 7 \bmod 9$ and $\left(\frac{3}{p}\right)_3 \neq 1$; then $A_F \cong A_K \cong \mathbb{Z}/3\mathbb{Z}$; see [4] or [6].

We give two consequences of Theorem 1.1. Let E_K be the group of units of K. Let E'_K be the subgroup of E_K generated by the units of non-trivial subfields of K. Write $q = [E_K : E_{K'}]$. One has [3, Theorem 12.1, 14.1]

$$q = 1 \text{ or } 3 \quad \text{and} \quad h_K = \frac{q}{3} h_F^2.$$

Here h_K (resp. h_F) is the class number of K (resp. F). Thus, if $p \equiv 4, 7 \bmod 9$ and $\left(\frac{3}{p}\right)_3 = 1$, Theorem 1.1 implies that $q = 3$. This confirms a conjecture made in [1].

Assume $p \equiv 4, 7 \bmod 9$. Theorem 1.1 implies that the norm equation $\mathbf{N}_{F/\mathbb{Q}}(x) = 3$ has a solution $x \in \mathcal{O}_F$ if and only if $\left(\frac{3}{p}\right)_3 = 1$, as mentioned in [12, p. 273]. Since $\mathcal{O}_F = \mathbb{Z}[\sqrt[3]{p}]$, this is to say, the Diophantine equation

$$x_1^3 + p x_2^3 + p^2 x_3^3 - 3 p x_1 x_2 x_3 = 3$$

has solutions $(x_1, x_2, x_3) \in \mathbb{Z}$ if and only if $\left(\frac{3}{p}\right)_3 = 1$.

2 The proof

2.1 Chevalley's ambiguous class number formula

We first review the S-version of Chevalley's ambiguous class number formula which will be used. For a finite set S of prime ideals of a number field L, the *S-class group* of K is defined as

$$\mathrm{Cl}_{K,S} := \mathrm{Cl}_K / \langle [\mathfrak{p}] : \mathfrak{p} \in S \rangle,$$

where Cl_K denotes the class group of K and $[\mathfrak{p}]$ denotes the ideal class of $\mathfrak{p}$. Let $E_{K,S} := \mathcal{O}_{K,S}^\times$ denote the group of S-units of K. Let L/K be a finite cyclic extension with Galois group G. For a finite set S of prime ideals of K, we denote by $\mathrm{Cl}_{L,S} = \mathrm{Cl}_{L,S_L}$ for simplicity, where S_L is the set of primes of L lying above those in S. Chevalley's ambiguous class number formula states that the order of the G-invariant subgroup of $\mathrm{Cl}_{L,S}$ is given by

$$|\mathrm{Cl}_{L,S}^G| = |\mathrm{Cl}_{K,S}| \cdot \frac{\prod_{v \notin S} e_v \cdot \prod_{v \in S} e_v f_v}{[L : K] \cdot [E_{K,S} : E_{K,S} \cap \mathbf{N} L^\times]}. \tag{2.1}$$

Here the first product runs over all places of K not in S, e_v and f_v are the ramification index and the residue degree of v respectively, and $\mathbf{N} = \mathbf{N}_{L/K}$ is the norm map. For a proof of this formula, see [10] for example. The unit index in (2.1) can be computed by Hilbert symbols provided that L/K is a Kummer extension.

Proposition 2.1 *Let L/K be a cyclic extension of degree d and $\mu_d \subset K$. Then $L = K(\sqrt[d]{a})$ for some $a \in K$. Let* Ram *be the set of ramified places of K. Define*

$$\rho : \frac{E_{K,S}}{(E_{K,S})^d} \longrightarrow \prod_{v \in S \cup \mathrm{Ram}} \mu_d$$
$$x \longmapsto \left(\left(\frac{x, a}{v} \right)_d \right)_{v \in S \cup \mathrm{Ram}}.$$

Then the kernel of ρ is given by

$$\operatorname{Ker}\rho = \frac{E_{K,S} \cap \mathbf{N}L^{\times}}{(E_{K,S})^d}$$

and hence the size of the image is given by

$$|\operatorname{Im}(\rho)| = [E_{K,S} : E_{K,S} \cap \mathbf{N}L^{\times}],$$

which is at most $d^{|S \cup \mathrm{Ram}|-1}$.

Proof This result is a standard direct consequence of local class field theory, Hasse's norm theorem, and the product formula for Hilbert symbols. For details, see [9, §2].

If $\sigma \in \operatorname{Aut}(K)$ and v is a prime of K, we have (*loc. cit.*)

$$\sigma\left(\frac{a,b}{v}\right)_d = \left(\frac{\sigma(a),\sigma(b)}{\sigma(v)}\right)_d, \quad a,b \in K^{\times}. \tag{2.2}$$

For our applications, the degree $[L:K]$ is a power of a prime ℓ. For any finite generated abelian group A, we denote by $A_\ell = A \otimes \mathbb{Z}_\ell$ where $\mathbb{Z}_\ell$ is the ring of ℓ-adic integers. If A is finite, A_ℓ is the ℓ-primary subgroup of A. If there is no ambiguity, we write a for $a \otimes 1 \in A_\ell$ for $a \in A$. Clearly, the formula (2.1) still holds by replacing $(\mathrm{Cl}_{L,S})^G$ and $\mathrm{Cl}_{K,S}$ with $\big((\mathrm{Cl}_{L,S})_\ell\big)^G = (\mathrm{Cl}_{L,S}^G)_\ell$ and $(\mathrm{Cl}_{K,S})_\ell$ respectively.

The following well known fact which is proved by counting the orbits of the G-action or by Nakayama's Lemma will be used frequently:

$$(\mathrm{Cl}_{L,S})_\ell = 0 \text{ if and only if } (\mathrm{Cl}_{L,S}^G)_\ell = 0.$$

2.2 Proof of Theorem 1.1

From now on, assume $p \equiv 1 \bmod 3$. Denote by

- $k = \mathbb{Q}(\mu_3)$;
- M^+ the unique cubic subfield of $\mathbb{Q}(\mu_p)$, which is real;
- $M = M(\mu_3)$ a quadratic extension of M^+;
- $L = KM = M(\sqrt[3]{p}, \mu_3)$;
- $A_T = (\mathrm{Cl}_T)_3$ for any number field T.

$$\begin{tikzcd}
 & L = KM \arrow[dl, dash] \arrow[d, dash] \arrow[dr, dash] & \\
K = \mathbb{Q}(\sqrt[3]{p}, \mu_3) \arrow[d, dash] \arrow[dr, dash] & FM^+ \arrow[dl, dash] \arrow[dr, dash] & M = kM^+ \arrow[dl, dash] \arrow[d, dash] \\
F = \mathbb{Q}(\sqrt[3]{p}) \arrow[dr, dash] & k = \mathbb{Q}(\mu_3) \arrow[d, dash] & M^+ \arrow[dl, dash] \\
 & \mathbb{Q} &
\end{tikzcd}$$

Proposition 2.2 *Assume $p \equiv 1 \bmod 3$.*

1. *There exists* $\alpha \in \mathcal{O}_k$ *such that* $M = k(\sqrt[3]{p\alpha})$ *and* $p = \alpha\overline{\alpha}$*;*
2. $A_M = 0$ *if and only if* $p \equiv 4, 7 \bmod 9$.

Proof 1. Since $p \equiv 1 \bmod 3$, we can write $p = \alpha\overline{\alpha}$ for some $\alpha \in \mathcal{O}_k$. Note that (α) and $(\overline{\alpha})$ are exactly the ramified primes of k in M. Now, since the class number of k is 1 and M/k is a Kummer extension, we have

$$M = \mathbb{Q}(\sqrt[3]{\gamma}) \quad \text{and} \quad \gamma = \zeta_3^a \alpha^b \overline{\alpha}^c$$

with $a, b, c \in \{0, 1, 2\}$ and $bc \neq 0$. Since $(3-a, 3-b, 3-c)$ gives the same field as (a, b, c), we conclude that $M = k(\sqrt[3]{\zeta_3^a p})$ or $k(\sqrt[3]{\zeta_3^a p\alpha})$ for some $a = 0, 1, 2$. Since M is abelian over $\mathbb{Q}$ but $k(\sqrt[3]{\zeta_3^a p})/\mathbb{Q}$ is not, M must coincide with $k(\sqrt[3]{\zeta_3^a p\alpha})$. By replacing α with $\zeta_3^a\alpha$, we have $M = k(\sqrt[3]{p\alpha})$ and $p = \alpha\bar{\alpha}$. This proves (1).

2. We apply (2.1) and Proposition 2.1 to the cyclic cubic extension M/k. Let $\iota : k \hookrightarrow \mathbb{Q}_p$ be the embedding induced by (α). Then we have the following equalities of cubic Hilbert symbols:

$$\left(\frac{\zeta_3, p\alpha}{\alpha}\right) = \iota^{-1}\left(\frac{\iota(\zeta_3), p\iota(\alpha)}{\mathbb{Q}_p}\right) = \iota^{-1}\left(\frac{\iota(\zeta_3), \iota(\alpha)}{\mathbb{Q}_p}\right)^{-1} = \zeta_3^{(p-1)/3}. \tag{2.3}$$

Hence this symbol as well as the index $[E_k : E_k \cap \mathbf{N}M^\times]$ is trivial if and only if $p \equiv 1 \bmod 9$. Thus

$$|A_M^G| = \frac{3^2}{3 \cdot [E_k : E_k \cap \mathbf{N}M^\times]} = 1$$

if and only if $p \equiv 4, 7 \bmod 9$. By Nakayama's lemma, it turns out that A_M is trivial if and only if $p \equiv 4, 7 \bmod 9$. This completes the proof of Proposition 2.2. □

Let $\mathfrak{p}$ (resp. $\mathfrak{p}'$) be the unique prime of M (resp. K) lying above $\alpha\mathcal{O}_k$. Then $\alpha\mathcal{O}_M = \mathfrak{p}^3$ and $\alpha\mathcal{O}_K = \mathfrak{p}'^3$.

Proposition 2.3 *Assume that* $p \equiv 1 \bmod 3$ *and* $\left(\frac{3}{p}\right)_3 = 1$.

1. *The extensions* L/K *and* FM^+/F *are both abelian unramified cubic extensions.*
2. *The primes* $\mathfrak{p}$ *and* $\mathfrak{p}'$ *both split in* L.

Proof 1. Since $L = KM^+$, we have that L/K is unramified outside the primes above p. Denote by $I_{(\alpha)}$ the inertia group of $(\alpha) = \alpha\mathcal{O}_k$ in the abelian extension L/k. By local class field theory and noting that the completion of k at (α) is $\mathbb{Q}_p$, we have a surjection

$$\mathbb{Z}_p^\times \twoheadrightarrow I_{(\alpha)}.$$

It follows that $I_{(\alpha)}$ can not be $\mathrm{Gal}(L/k) \cong (\mathbb{Z}/3\mathbb{Z})^2$. On the other hand, $I_{(\alpha)}$ is non-trivial since (α) is ramified in K and M. This shows that $\mathfrak{p}$ and $\mathfrak{p}'$ must be unramified in L. An entirely similar argument for the prime $(\overline{\alpha}) = \overline{\alpha}\mathcal{O}_k$ shows that L/M and L/K are both unramified outside the primes above $\overline{\alpha}$. This shows that L/K is unramified everywhere.

For the extension FM^+/F, first note that it is unramified outside $\sqrt[3]{p}\mathcal{O}_F$ as $M^+/\mathbb{Q}$ is unramified outside p. We claim that $\sqrt[3]{p}\mathcal{O}_F$ is also unramified in FM^+. Otherwise, since K/F is unramified at $\sqrt[3]{p}\mathcal{O}_F$, the prime of K above $\sqrt[3]{p}$ would be ramified in L. But this contradicts that L/K is unramified whence the claim holds. This proves (1).

2. We have just shown that FM^+ is contained in the Hilbert class field of F. By class field theory, the principal prime $\sqrt[3]{p}\mathcal{O}_F$ splits in FM^+. It follows that $\mathfrak{p}'$ and $\mathfrak{p}$ both split in L. □

Lemma 2.4 1. *If $p \equiv 4, 7 \bmod 9$, then 3 is totally ramified in K.*
2. *If $p \equiv 1 \bmod 3$ and 3 is a cube modulo p, then $(1-\zeta_3)\mathcal{O}_k$ splits in M.*

Proof 1. Since $(x+p)^3 - p$ is an Eisenstein polynomial, 3 is totally ramified in F. Since 3 is also ramified in k, it follows that 3 is totally ramified in K by counting the ramification degrees.

2. Fix the canonical isomorphism

$$\begin{aligned}(\mathbb{Z}/p\mathbb{Z})^\times &\cong \mathrm{Gal}(\mathbb{Q}(\mu_p)/\mathbb{Q})\\ a &\mapsto (\sigma_a : \zeta_p \mapsto \zeta_p^a).\end{aligned}$$

By definition, M^+ is the subfield of $\mathbb{Q}(\mu_p)$ fixed by $(\mathbb{Z}/p\mathbb{Z})^{\times 3}$. Our assumptions imply that σ_3 is trivial on M^+ whence 3 splits in M^+. It follows that $(1-\zeta_3)\mathcal{O}_k$ must split in M. □

We need the following elementary fact on the local field $\mathbb{Q}_3(\mu_3)$.

Lemma 2.5 *If $a, b \in \mathbb{Z}$ with $3 \nmid ab$, then the cubic Hilbert symbol of a and b in $\mathbb{Q}_3(\mu_3)$ is trivial.*

Proof By convergence of the Taylor expansion of $(1+9x)^{1/3}$ on $\mathbb{Z}_3[\mu_3]$, every element in $1+9\mathbb{Z}_3[\mu_3]$ is a cube. Note that -1 is a cube whence the cubic Hilbert symbol $\left(\frac{a,a}{\mathbb{Q}_3(\mu_3)}\right) = 1$. Thus, we only need to show the triviality of the symbol

$$\left(\frac{4,7}{\mathbb{Q}_3(\zeta_3)}\right) = \left(\frac{4,2}{\mathbb{Q}_3(\zeta_3)}\right) = \left(\frac{2,2}{\mathbb{Q}_3(\zeta_3)}\right)^2 = 1.$$

□

Theorem 2.6 *Assume that $p \equiv 4, 7 \bmod 9$ and $\left(\frac{3}{p}\right)_3 = 1$. Then A_L is non-trivial and 3 does not divides $|\mathrm{Cl}_{L,\{\mathfrak{p}\}}|$.*

Proof We first apply (2.1) on L/M with $S = \emptyset$ to prove 3 divides $|A_L^G|$ where $G = \mathrm{Gal}(L/M)$. By Proposition 2.3 and Lemma 2.4, exactly the three primes $\mathfrak{l}, \sigma(\mathfrak{l}), \sigma^2(\mathfrak{l})$ of M lying above $(1-\zeta_3)\mathcal{O}_k$ are ramified in L/M, where σ is a generator of $\mathrm{Gal}(M/k)$. By Proposition 2.2, we know that $|A_M| = 1$. It remains to compute the unit index. Note that $L = M(\sqrt[3]{p})$. To apply Proposition 2.1, we define

$$\begin{aligned}\rho : E_M &\longrightarrow \mu_3^3\\ u &\longmapsto \left(\left(\frac{u,p}{\mathfrak{l}}\right), \left(\frac{u,p}{\sigma(\mathfrak{l})}\right), \left(\frac{u,p}{\sigma^2(\mathfrak{l})}\right)\right).\end{aligned}$$

Since M/M^+ is a CM-extension, the group $(E_M)_3$ is generated by $(E_{M^+})_3$ and ζ_3 by [11, Theorem 4.12]. The completion of M^+ at a prime above 3 is $\mathbb{Q}_3$. It follows that $\eta \equiv a \bmod 9$ with $a \in \mathbb{Z}$ for any $\eta \in E_{M^+}$. Then by Lemma 2.5,

$$|\rho(E_{M^+})| = 1. \tag{2.4}$$

Now we compute $\rho(\zeta_3)$. Since $\sigma(\zeta_3) = \zeta_3$, by (2.2) we have

$$\left(\frac{\zeta_3, p}{\mathfrak{l}}\right) = \left(\frac{\zeta_3, p}{\sigma(\mathfrak{l})}\right) = \left(\frac{\zeta_3, p}{\sigma^2(\mathfrak{l})}\right).$$

By Lemma 2.4, the completion of M at $\mathfrak{l}$ is $\mathbb{Q}_3(\mu_3)$. Applying the product formula for cubic Hilbert symbols on the field $\mathbb{Q}(\mu_3)$ gives

$$\left(\frac{\zeta_3, p}{(\alpha)}\right)\left(\frac{\zeta_3, p}{(\overline{\alpha})}\right)\left(\frac{\zeta_3, p}{(1-\zeta_3)}\right) = 1.$$

By (2.3) and our assumption $p \equiv 4, 7 \bmod 9$, we obtain that

$$\left(\frac{\zeta_3, p}{(1-\zeta_3)}\right) \neq 1 \text{ and } \left(\frac{\zeta_3, p}{\mathbb{Q}_3(\mu_3)}\right) \neq 1. \tag{2.5}$$

This proves that $\rho(\zeta_3) = \zeta_3^{\pm 1}(1, 1, 1)$. Combining with (2.4), we conclude that $|\rho(E_M)_3| = 3$. Then Chevalley's formula gives

$$|A_L^G| = \frac{3^3}{3 \times 3} = 3.$$

In particular, $|A_L| \geq 3$.

Next, we apply Chevalley's formula on L/M with $S = \{\mathfrak{p}\}$ to compute $\mathrm{Cl}_{L,\{\mathfrak{p}\}}^G$. Define

$$\beta = \frac{\sqrt[3]{p\alpha}}{\mathbf{N}_{\mathbb{Q}(\mu_p)/M^+}(1-\zeta_p)}.$$

Note that β^3 generates the ideal $\alpha\mathcal{O}_M$ whence $\beta\mathcal{O}_M = \mathfrak{p}$. It follows that $(E_{M,\{\mathfrak{p}\}})_3$ is generated by β, ζ_3 and E_{M^+}. We claim that

$$\left(\frac{\beta, p}{\mathfrak{l}}\right) \neq \left(\frac{\beta, p}{\sigma(\mathfrak{l})}\right).$$

Indeed, by (2.2), the right hand side equals the Hilbert symbol of $\sigma^{-1}(\beta)$ and p at $\mathfrak{l}$. Note that $\sigma^{-1}(\beta) = \zeta_3^{\pm 1}\beta\eta$ for some $\eta \in E_{M^+}$. Thus the inequality follows from (2.4) and (2.5). By Proposition 2.1, this shows that the index

$$[E_{M,\{\mathfrak{p}\}} : E_{M,\{\mathfrak{p}\}} \cap \mathbf{N}L^\times] = 9.$$

By Proposition 2.3, the prime $\mathfrak{p}$ splits in L. It follows from (2.1) that 3 does not divide $|\mathrm{Cl}_{L,\{\mathfrak{p}\}}^G|$ whence 3 does not divide $|\mathrm{Cl}_{L,\{\mathfrak{p}\}}|$ by Nakayama's Lemma. This completes the proof. □

Proof of Theorem 1.1 By Theorem 2.6, A_L is non-trivial. It follows that, by Nakayama's lemma, we have $|A_L^{\mathrm{Gal}(L/K)}| \geq 3$. Since L/K is unramified everywhere, by Hasse's norm theorem and local class field theory (or Proposition 2.3), we have the unit index $[E_K : E_K \cap \mathbf{N}(L^\times)] = 1$. Then applying Chevalley's formula with $S = \emptyset$ to the extension L/K gives

$$|A_K| \geq 9.$$

Recall that $\mathfrak{p}'$ is the prime of K lying above $\alpha\mathcal{O}_k$. Note that $\mathfrak{p}'\mathcal{O}_L = \mathfrak{p}\mathcal{O}_L$, we have $\mathrm{Cl}_{L,\{\mathfrak{p}'\}} = \mathrm{Cl}_{L,\{\mathfrak{p}\}}$ by definition. Since 3 does not divide $|\mathrm{Cl}_{L,\{\mathfrak{p}'\}}|$ by Theorem 2.6 and $\mathfrak{p}'$ splits in L by Proposition 2.3, Chevalley's formula with $S = \{\mathfrak{p}'\}$ will imply that $(\mathrm{Cl}_{K,\{\mathfrak{p}'\}})_3 \cong \mathbb{Z}/3\mathbb{Z}$ if we can show that

$$[E_{K,\{\mathfrak{p}'\}} : E_{K,\{\mathfrak{p}'\}} \cap \mathbf{N}(L^\times)] = 1.$$

Because $\mathfrak{p}'$ splits in L/K by Proposition 2.3, the local extension at $\mathfrak{p}'$ is trivial. Thus any $\{\mathfrak{p}'\}$-unit is a local norm at $\mathfrak{p}'$ whence is a local norm at every place of K as L/K is unramified. By Hasse's norm theorem, the above unit index is indeed trivial.

The equality $\mathfrak{p}'^3 = \alpha\mathcal{O}_K$ implies that $|\mathrm{Cl}_K| \leq 3|\mathrm{Cl}_{K,\{\mathfrak{p}'\}}|$. It follows that

$$|A_K| \leq 9.$$

Hence $|A_K| = 9$ and then $|A_F| = 3$ by [7, Lemma 1].

Let τ be the non-trivial element of $\Delta = \mathrm{Gal}(K/F)$. Since Δ is of order 2, we have a decomposition of $\mathbb{Z}_3[\Delta]$-modules

$$A_K = A_K^+ \oplus A_K^-, \text{ where } A_K^{\pm} = \{a \in A_K | \tau(a) = a^{\pm 1}\}.$$

It is well known that $|A_K^+| = |A_F| = 3$ (for example, using (2.1)). Thus A_K has a direct factor $\mathbb{Z}/3\mathbb{Z}$. This implies that $A_K \cong (\mathbb{Z}/3\mathbb{Z})^2$, completing the proof of Theorem 1.1. □

Remark 2.7 René Schoof informed us a refinement of Theorem 2.6 based on our results as follows. In which case, one can derive that $|A_L^H| = 3$ for any proper subgroup H of $\mathrm{Gal}(L/k) \cong (\mathbb{Z}/3\mathbb{Z})^2$, by Chevalley's formula and Theorem 1.1. Then it can be shown that $A_L \cong \mathbb{Z}/3\mathbb{Z}$ by a purely algebraic result on modules over $\mathbb{Z}_3[\mathrm{Gal}(L/k)]$.

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